A Random Walk Through Lantern Dynamics

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Demo deck note: all names, events, and scientific claims are placeholders.

Outline

1. Setting
2. Figure
3. Math
4. Logic
5. Wrap-Up

Setting

Why Lanterns?

A harmless fictional motivation

Toy problem

Suppose a lantern drifts through a quiet city square while its brightness reacts to wind, memory, and moonlight. We ask whether the observed glow can reveal the hidden drift pattern.

- The state variable is the latent position x_t.
- The observable is the brightness signal b_t.
- The nuisance parameter is the evening breeze η_t.

Main message

A clean slide should tell the audience what to look at before asking them to interpret details.

Template purpose

This content is intentionally random, so the layout can be reused for any real topic.

Ingredients

Minimal objects for the story

1. A hidden trajectory $x_0, x_1, \ldots, x_T$ moving through a two-dimensional plaza.
2. A response function $g(x_t)$ that converts location into observed brightness.
3. A correction term that remembers whether the lantern was bright a moment ago.
4. A small amount of noise, because perfectly behaved lanterns are suspicious.

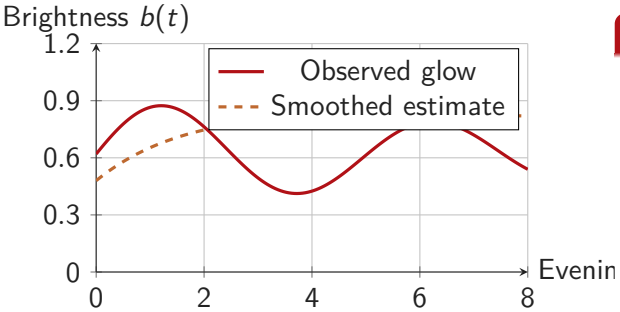
Example sentence

"The lantern forgets slowly, drifts politely, and reports its location through light."

Figure

Figure with Explanation

Two invented brightness curves



Reading the figure

- The observed glow oscillates early in the evening.
- The smoothed estimate rises more calmly.
- The two curves agree once the transient behavior fades.

Reusable pattern

Pair the plot with a short reading guide.

Math

Pure Math Example

A compact fictional model

Latent process

For position $x_t \in \mathbb{R}^2$ and noise ξ_t,

$$x_{t+1} = x_t + \alpha \begin{pmatrix} \cos\theta_t \\ \sin\theta_t \end{pmatrix} + \sigma\xi_t.$$

Brightness functional

$$b_t = \exp\left(-\frac{\|x_t - c\|^2}{2\rho^2}\right) + \lambda b_{t-1} + \varepsilon_t,$$

Here c is the plaza center and λ controls memory.

Interpretation

Logic

Derivation Logic

Memory parameter

Goal. Estimate the memory coefficient λ in

$$b_t = z_t + \lambda b_{t-1} + \varepsilon_t, \qquad z_t \text{ known.}$$

$$r_t = b_t - z_t,$$

$$J(\lambda) = \sum_{t=1}^{T} (r_t - \lambda b_{t-1})^2,$$

$$0 = -2 \sum_{t=1}^{T} b_{t-1}(r_t - \lambda b_{t-1}),$$

$$\hat{\lambda} = \frac{\sum_{t=1}^{T} b_{t-1} r_t}{\sum_{t=1}^{T} b_{t-1}^2}.$$

Derivation Logic

How to present the chain

Make explicit

- What is observed
- What is assumed known
- Which quantity is being estimated

Keep short

- Avoid algebra that does not change the idea.
- Use one equation per logical step.
- Move edge cases to backup slides.

Speaking note

The audience should feel the derivation moving forward, not just expanding sideways.

Wrap-Up

Summary

What this random demo illustrates

- Metadata fields for title, subtitle, authors, institute, date, conference, and title note
- A title page using the closing-style background
- Content pages using the CityUHK content background
- Automatic section pages using the section background
- Example slides for bullets, figures, equations, and derivations

Next step

Replace the lantern story with your actual research, teaching material, or project update.

Thank You

Questions and comments